

5 (a) Pigs are farm animals used for livestock in some parts of the world. The first genetically modified (transgenic) pigs were produced in 1985. Foreign DNA was injected directly into the nuclei of zygotes. The foreign DNA was made up of two components:

- the gene coding for human growth hormone
- a section of mouse DNA that, in the presence of metal ions, allows transcription to begin.

(i) The human growth hormone synthesised by the transgenic pigs had the effect of making the pigs grow faster, larger and heavier than non-genetically modified pigs.

Suggest reasons for this difference.

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(ii) Suggest **and** explain why the mouse DNA was included in the foreign DNA.

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Only 1% of the attempts successfully produced transgenic pigs. These pigs showed higher body mass and a greater muscle to fat ratio than normal pigs. Monitoring of the pigs' behaviour revealed that they rested more than normal pigs, suffered from stomach ulcers and were unwilling to mate.

(b) (i) Discuss whether these transgenic pigs have long term economic value.

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(ii) Comment on the ethics of producing transgenic pigs showing the features described.

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(c) In 2015 pigs were produced that had part of their genome altered by a new technique. The technique involved:

- an RNA sequence designed to bind to a specific targeted pig gene
- a gene-editing enzyme that is able to cut out sections of DNA.

The technique was used on pig zygotes that had been created by IVF. All the zygotes treated grew into piglets and these all showed large deletions in the targeted gene.

This gene coded for a specific cell surface membrane protein. The piglets did not express the protein and this gave them resistance to infection by a virus that causes a serious disease in pigs.

(i) Describe **two** advantages of the gene-editing technique compared to the traditional genetic modification technique used to make transgenic pigs in 1985.

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(ii) A scientist stated that this new technique is a form of selective breeding, so is not genetic engineering.

Discuss whether this statement is true **and** whether public groups who oppose transgenic animals will be more or less likely to accept the new technique.

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